Projected Gradient Descent for Smooth Convex Optimization

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1 Algorithm Description

The algorithm iteratively updates parameters by taking gradient steps and projecting the result back onto the constraint set at each iteration. It is designed for smooth convex optimization problems with closed, convex, and bounded constraint domains.

1.1 Mathematical Formulation

Problem Statement: Given a smooth convex function $f : \mathbb{R}^d \to \mathbb{R}$ with L-Lipschitz continuous gradient (i.e., $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x, y), and a closed, convex, and bounded constraint domain $\mathcal{C} \subset \mathbb{R}^d$, find:

$$w^* = \arg\min_{w \in \mathcal{C}} f(w)$$

Update Rules and Equations:

1. **Gradient Step**: Compute the gradient of f at current parameters w_t:

$$g_t = \nabla f(w_t)$$

2. **Parameter Update (Gradient Step)**:

$$w_{t+1/2} = w_t - \eta_t g_t$$

where η_t is the step size at iteration t.

3. **Projection onto Constraint Set**:

$$w_{t+1} = \Pi_{\mathcal{C}}(w_{t+1/2})$$

where $\Pi_{\mathcal{C}}(x) = \arg\min_{y \in \mathcal{C}} \|x - y\|$ is the projection operator onto $\mathcal{C}$.

Hyperparameters and Their Roles:

- η_t: Step size at iteration t. Can be constant or adaptive. Controls the aggressiveness of each update.
- T: Total number of iterations. Determines the algorithm's computational budget.
- $\mathcal{C}$: Constraint domain. Defines the feasible region for the solution.

2 Pseudocode and Dry Run

2.1 Pseudocode

2.2 Dry Run Example

Problem Setup:

Algorithm 1 Projected Gradient Descent (PGD)

Input:
Smooth convex function f with gradient ∇f
Constraint domain $\mathcal{C}$
Initial guess w_0
Step size schedule $\{\eta_t\}_{t=0}^{T-1}$
Total iterations T
Initialize:
$t = 0$
$w_t = w_0$
while $t < T$ **do**
Compute Gradient: $g_t = \nabla f(w_t)$
Gradient Step: $w_{t+1/2} = w_t - \eta_t g_t$
Project onto Constraint Set: $w_{t+1} = \Pi_{\mathcal{C}}(w_{t+1/2})$
Update Iteration: $t = t + 1$
end while
Return w_T as the approximate solution

- **Function**: $f(w) = (w - 3)^2$
- **Constraint Domain**: $\mathcal{C} = [0, 4]$ (closed, convex, and bounded)
- **Initialization**: $w_0 = 0$
- **Step Size**: $\eta = 0.1$ (constant for simplicity)
- **Minibatch Size**: 1 (deterministic, so no sampling variability)
- **Iterations to Show**: 3

2.2.1 Iteration 1

1. **Gradient Computation**:

$$g_0 = \nabla f(w_0) = 2(w_0 - 3) = 2(0 - 3) = -6$$

2. **Parameter Update**:

$$w_{1/2} = w_0 - \eta g_0 = 0 - 0.1(-6) = 0.6$$
$$w_1 = \Pi_{[0,4]}(0.6) = 0.6 \quad (\text{since } 0.6 \in [0, 4])$$

3. **Objective Value**:

$$f(w_1) = (0.6 - 3)^2 = (-2.4)^2 = 5.76$$

2.2.2 Iteration 2

1. **Gradient Computation**:

$$g_1 = \nabla f(w_1) = 2(0.6 - 3) = -4.8$$

2. **Parameter Update**:

$$w_{2/2} = w_1 - \eta g_1 = 0.6 - 0.1(-4.8) = 0.6 + 0.48 = 1.08$$

$$w_2 = \Pi_{[0,4]}(1.08) = 1.08$$

3. **Objective Value**:

$$f(w_2) = (1.08 - 3)^2 = (-1.92)^2 = 3.6864$$

2.2.3 Iteration 3

1. **Gradient Computation**:
$$g_2 = \nabla f(w_2) = 2(1.08 - 3) = -3.84$$

2. **Parameter Update**:
$$w_{3/2} = w_2 - \eta g_2 = 1.08 - 0.1(-3.84) = 1.08 + 0.384 = 1.464$$
$$w_3 = \Pi_{[0,4]}(1.464) = 1.464$$

3. **Objective Value**:
$$f(w_3) = (1.464 - 3)^2 = (-1.536)^2 = 2.3593$$

2.2.4 Dry Run Summary

Iteration	w_t	g_t	$f(w_t)$
0	0	-6	9
1	0.6	-4.8	5.76
2	1.08	-3.84	3.6864
3	1.464		2.3593

Table 1: Summary of Dry Run Iterations

3 Convergence Theory

3.1 Assumptions

3.1.1 Formal Statement of Assumptions

1. **Smoothness**: The objective function $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth, i.e., $\forall x, y \in \mathcal{C}$,
$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$$
where $\mathcal{C} \subseteq \mathbb{R}^n$ is the constraint set, and $L > 0$.

2. **Convexity**: f is convex, i.e., $\forall x, y \in \mathcal{C}$ and $\lambda \in [0, 1]$,
$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

3. **Constraint Set**: $\mathcal{C}$ is closed, convex, and bounded with diameter D, i.e., $\sup_{x,y\in\mathcal{C}} \|x - y\| \leq D$.

4. **Parameter Constraints**: Step size η satisfies $0 < \eta \leq \frac{1}{L}$.

5. **Function Properties**: Let $f^* = \min_{x\in\mathcal{C}} f(x)$, assuming f^* is attainable at some $x^* \in \mathcal{C}$.

3.2 Main Convergence Theorem

3.2.1 Convergence Theorems for Gradient Projection Algorithm

Theorem 1: Rate of Convergence

Given the assumptions above, for the sequence $\{x_t\}$ generated by the Gradient Projection Algorithm with step size $\eta = \frac{1}{L}$,
$$f\left(\frac{1}{T}\sum_{t=1}^{T} x_t\right) - f^* = O\left(\frac{1}{T}\right)$$

as $T \to \infty$.

Theorem 2: Expected Reduction Properties

For any t,

$$\mathbb{E}[f(x_{t+1})] - f(x_t) \leq -\eta \|\nabla f(x_t)\|^2 + \frac{L\eta^2}{2} \|\nabla f(x_t)\|^2$$

Given $\eta \leq \frac{1}{L}$, this implies an expected reduction in function value at each step.

Theorem 3: Special Case Behaviors

- **Linear Convergence for Strongly Convex Functions**: If f is μ-strongly convex ($\mu > 0$), the algorithm achieves linear convergence, i.e.,

$$f(x_t) - f^* \leq (1 - \eta\mu)^t (f(x_0) - f^*)$$

  with $\eta = \frac{1}{L}$.

3.3 Theoretical Properties

3.3.1 Key Theoretical Properties

1. Stability Analysis

The algorithm's stability can be analyzed through the lens of its sensitivity to initial conditions and perturbations. Given the projection step ensures $x_t \in \mathcal{C}$ for all t, the sequence remains bounded.

2. Sensitivity to Hyperparameters

- **Step Size** η: The choice of η significantly affects convergence. $\eta = \frac{1}{L}$ balances the trade-off between step size and smoothness constant for optimal convergence in smooth convex problems.
- **Initial Point** x_0: While the convergence rate is largely independent of x_0 due to the convexity and boundedness assumptions, the initial point can influence the total number of steps required to reach a certain accuracy level in practice.

3. Comparison to Baseline Methods

- **Unconstrained Gradient Descent**: The Gradient Projection Algorithm generalizes unconstrained GD by adding a projection step, crucial for handling constraints.
- **Interior Point Methods**: Unlike interior point methods, this algorithm does not require the constraint set to have a non-empty interior or to solve complex sub-problems at each iteration, making it more efficient for certain types of constraints.

3.4 Discussion

The developed convergence theory for the Gradient Projection Algorithm on smooth convex optimization problems with closed, convex, and bounded constraint domains provides insights into its efficiency and behavior. The $O(1/T)$ rate of convergence positions it favorably among first-order methods for such problems. Linear convergence for strongly convex objectives further highlights its versatility. Sensitivity analysis suggests careful selection of hyperparameters, particularly the step size, for optimal performance. Comparison with baseline methods underscores its simplicity and efficiency, especially in scenarios where constraints are complex but projection can be efficiently computed.

4 Convergence Proof

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

- **Gradient Projection Algorithm Update Rule**:

$$x_{t+1} = \Pi_{\mathcal{C}} (x_t - \eta \nabla f(x_t))$$

where $\Pi_{\mathcal{C}}(x)$ is the projection of x onto $\mathcal{C}$.

- **Projection Property**: For any $x \in \mathbb{R}^n$ and $y \in \mathcal{C}$,

$$\langle x - \Pi_{\mathcal{C}}(x), y - \Pi_{\mathcal{C}}(x) \rangle \leq 0$$

4.1.2 Lemmas

Lemma 1: Smoothness Implies Descent
Given f is L-smooth,

$$f(x_{t+1}) \leq f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{L}{2} \|x_{t+1} - x_t\|^2$$

Lemma 2: Projection Inequality
For $x_{t+1} = \Pi_{\mathcal{C}}(x_t - \eta \nabla f(x_t))$ and any $x^* \in \mathcal{C}$,

$$\|x_{t+1} - x^*\|^2 \leq \|x_t - \eta \nabla f(x_t) - x^*\|^2$$

4.2 Step-by-Step Proof

Goal: Derive $f\left(\frac{1}{T}\sum_{t=1}^{T} x_t\right) - f^* = O\left(\frac{1}{T}\right)$

4.2.1 Step 1: Apply Smoothness (Lemma 1)

Starting with Lemma 1 and substituting x_{t+1}:

$$f(x_{t+1}) \leq f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{L}{2} \|x_{t+1} - x_t\|^2$$

4.2.2 Step 2: Utilize Projection Inequality (Lemma 2)

For $x^* \in \mathcal{C}$:

$$\|x_{t+1} - x^*\|^2 \leq \|x_t - \eta \nabla f(x_t) - x^*\|^2$$

Expanding both sides:

$$\|x_{t+1}\|^2 - 2\langle x_{t+1}, x^* \rangle + \|x^*\|^2 \leq \|x_t\|^2 - 2\eta \langle \nabla f(x_t), x_t - x^* \rangle + \eta^2 \|\nabla f(x_t)\|^2 + \|x^*\|^2$$

Simplifying and rearranging gives:

$$\langle \nabla f(x_t), x_t - x^* \rangle \leq \frac{1}{2\eta}\left(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2\right) + \frac{\eta}{2} \|\nabla f(x_t)\|^2$$

4.2.3 Step 3: Convexity and Telescoping Sum

By convexity, $f(x_t) - f(x^*) \leq \langle \nabla f(x_t), x_t - x^* \rangle$. Substituting from Step 2:

$$f(x_t) - f(x^*) \leq \frac{1}{2\eta}\left(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2\right) + \frac{\eta}{2} \|\nabla f(x_t)\|^2$$

Summing over $t = 1$ to T:

$$\sum_{t=1}^{T} (f(x_t) - f(x^*)) \leq \frac{1}{2\eta}\left(\|x_1 - x^*\|^2 - \|x_{T+1} - x^*\|^2\right) + \frac{\eta}{2} \sum_{t=1}^{T} \|\nabla f(x_t)\|^2$$

Since $\|x_{T+1} - x^*\|^2 \geq 0$ and using the boundedness of $\mathcal{C}$ ($\|x_1 - x^*\| \leq D$):

$$\sum_{t=1}^{T} (f(x_t) - f(x^*)) \leq \frac{D^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \|\nabla f(x_t)\|^2$$

4.2.4 Step 4: Average Function Value

Define $\bar{x}_T = \frac{1}{T}\sum_{t=1}^{T} x_t$. By Jensen's Inequality (since f is convex):

$$f(\bar{x}_T) \le \frac{1}{T}\sum_{t=1}^{T} f(x_t)$$

Thus:

$$f(\bar{x}_T) - f(x^*) \le \frac{1}{T}\left(\frac{D^2}{2\eta} + \frac{\eta}{2}\sum_{t=1}^{T} \|\nabla f(x_t)\|^2\right)$$

4.2.5 Step 5: Bounding Gradient Norms

Assuming $\|\nabla f(x_t)\|$ is bounded by G for all t (which can be justified under the given assumptions for smooth and convex functions over a bounded domain):

$$\sum_{t=1}^{T} \|\nabla f(x_t)\|^2 \le TG^2$$

Substituting back:

$$f(\bar{x}_T) - f(x^*) \le \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}$$

Choosing $\eta = \frac{D}{G\sqrt{T}}$ (balancing the terms for optimal convergence):

$$f(\bar{x}_T) - f(x^*) \le \frac{GD}{2\sqrt{T}} + \frac{GD}{2\sqrt{T}} = \frac{GD}{\sqrt{T}}$$

Thus, $f(\bar{x}_T) - f(x^*) = O\left(\frac{1}{\sqrt{T}}\right)$ for this specific step size choice. However, to align with the requested $O(1/T)$ rate under the assumption $\eta = \frac{1}{L}$ (as in Theorem 1), we recognize an oversight in directly deriving $O(1/T)$ without additional assumptions (e.g., strong convexity for linear rate or specific step size schedules). The $O(1/T)$ claim in Theorem 1 suggests an expectation or average case analysis not fully elaborated here due to the step size choice and assumption strictness.

4.3 Rate Derivation (With Constants)

Given the misalignment in the direct derivation of $O(1/T)$ with constant step size $\eta = \frac{1}{L}$ without strong convexity, let's correct the course for a standard convergence rate under given assumptions:

- **For** $\eta = \frac{1}{L}$ (without strong convexity):

  $$f(\bar{x}_T) - f(x^*) = O\left(\frac{1}{\sqrt{T}}\right)$$

  This is because, with $\eta = \frac{1}{L}$, the dependence on T in the bound would typically reflect a $O\left(\frac{1}{\sqrt{T}}\right)$ rate for the average function value in stochastic or non-strongly convex settings, acknowledging the previous derivation's step size choice was optimized for a different scenario.

4.4 Special Cases

4.4.1 Linear Convergence for Strongly Convex Functions

- **Assumption**: f is μ-strongly convex.
- **Update Rule Remains**: $x_{t+1} = \Pi_{\mathcal{C}}(x_t - \eta\nabla f(x_t))$

- **Key Inequality** (Strong Convexity):

$$f(x_{t+1}) - f(x^*) \leq (1 - \eta\mu)\,(f(x_t) - f(x^*)) + \frac{L\eta^2}{2}\|\nabla f(x_t)\|^2$$

- **Derivation** (Simplified): With $\eta = \frac{1}{L}$ and strong convexity, through iterative application and considering the projection does not increase the distance to x^* in a strongly convex setting:

$$f(x_t) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^t (f(x_0) - f(x^*))$$

  Linear Convergence:

$$f(x_t) - f(x^*) \leq (1 - \eta\mu)^t\,(f(x_0) - f^*)$$